

PORTFLOW SBX

Container congestion predictor & port operations optimiser — a 72-hour decision cockpit that forecasts hotspots, reroutes vessels, optimises berths & cranes, and writes the shift-by-shift operations plan.

72_h

FORECAST HORIZON

4

REAL ENGINES, ONE PIPELINE

R² 0.90

SCHEDULE-AWARE MODEL

Bob

LOAD-BEARING AI ASSISTANT

100 ships offshore. A supply chain on hold.

In 2021, San Pedro Bay — the busiest US container gateway — became a floating warehouse. Berth, crane and yard planning ran on spreadsheets, hotspots were discovered reactively, and routing was decided after vessels were already on final approach.

- **Spreadsheet berth planning** — manual assignments, no lookahead, contradictions found at the shift meeting.
- **Reactive hotspot discovery** — congestion is measured after the queue has already exploded.
- **Routing decided too late** — divert/slow-steam options priced when it is too late to act on them.

~109

QUEUE PEAK • JAN 2022

100+

SHIPS WAITING OFFSHORE

\$10B+

ESTIMATED SUPPLY-CHAIN IMPACT

SOURCES: SOUTHERN CALIFORNIA MARINE EXCHANGE QUEUE COUNTS • 2021 BACKLOG REPORTING

Real terminal facts. Honestly-labelled vessel data.

TERMINAL	PIER	BERTH LENGTH	CRANES	CAPACITY
LBCT	Pier E	4,200 ft	18 STS	3.5M+ TEU/yr
ITS	Pier G	4,250 ft	14	n/a
PCT	Pier J	5,902 ft	14	n/a
TTI	Pier T	5,000 ft	16	n/a

MODELLED: 13 WORKING BERTHS · 62 CRANES AFTER PUBLISHED SPLITS
PORT-WIDE POLB: 80 BERTHS · 10 PIERS · 71 POST-PANAMAX CRANES
SOURCES: POLB TERMINAL FACT SHEETS (CITED IN DOCS/)

REAL PIPELINE · LABELLED DEMO DEFAULT

The vessel queue & congestion history ship as a clearly-marked **DEMO_AIS** dataset — every record is labelled at the API and in the docs.

The replacement path is real and tested: `scripts/ais` turns a genuine **NOAA AccessAIS** export into the engine's exact series format — anchorage classification, per-MMSI dwell, hourly queue/wait aggregation, identical index formula.

build-congestion.ts

import-series.ts

source="AIS"

Schedule-aware ridge regression, 72-hour rollout.

0.90

R² · 48H
HOLDOUT

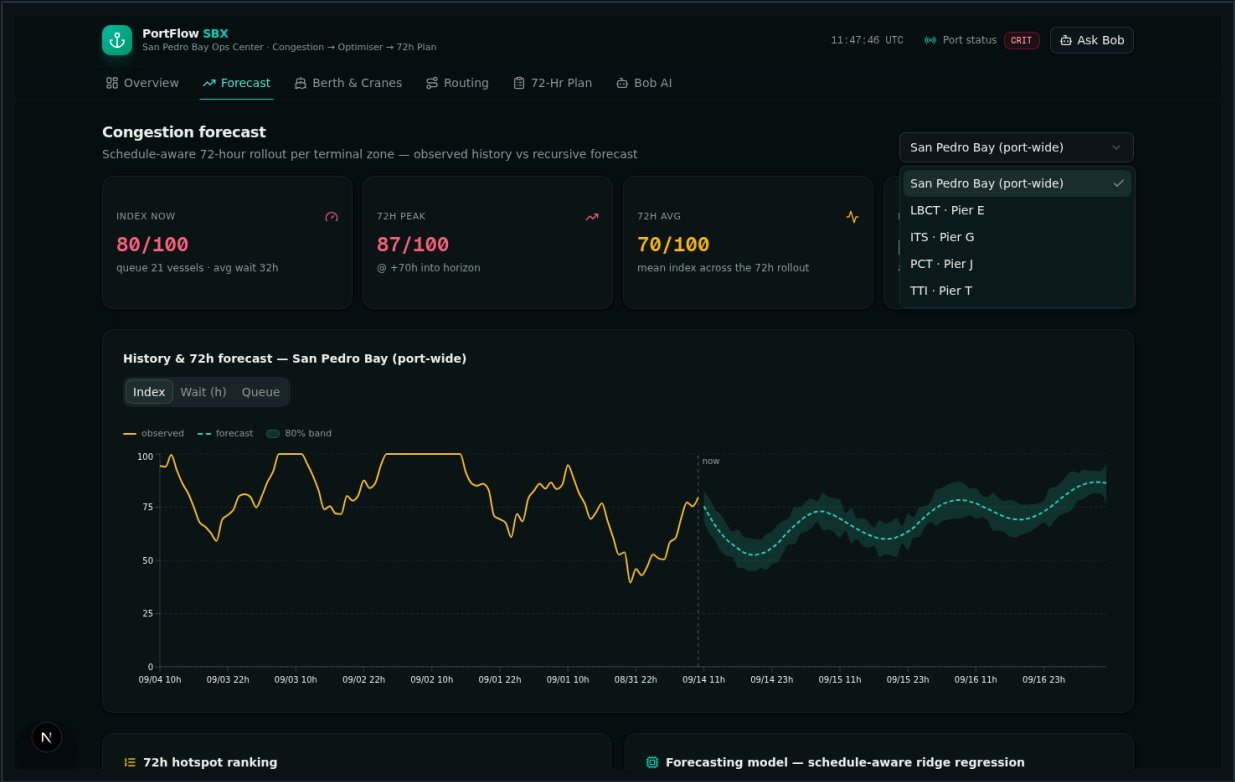
≈5

MAE @72H · IDX
PTS

+83%

SKILL VS
PERSISTENCE

- 01 **Schedule-aware features.** 14 inputs — inbound-ETA arrival pressure, berth load factor, diurnal terms, queue state, weather flag.
- 02 **Ridge regression.** Normal equations + Gaussian elimination, $\lambda = 3.0$ — trained per zone on hourly history.
- 03 **Recursive rollout.** 72 one-step forecasts with damped recursion ($\delta = 0.75$) — no flat-line saturation.
- 04 **Honest uncertainty.** 48h holdout + 12 multi-origin rollouts → per-horizon σ , 80% bands ($\pm 1.2816\sigma$), MAE & skill reported.



DASHBOARD · FORECAST TAB — HISTORY, 72H FORECAST, 80% BAND, MODEL CARD

Three phases beat FIFO — with honest trade-offs.

19

SERVICED (+1 VS
FIFO)

+10.4%

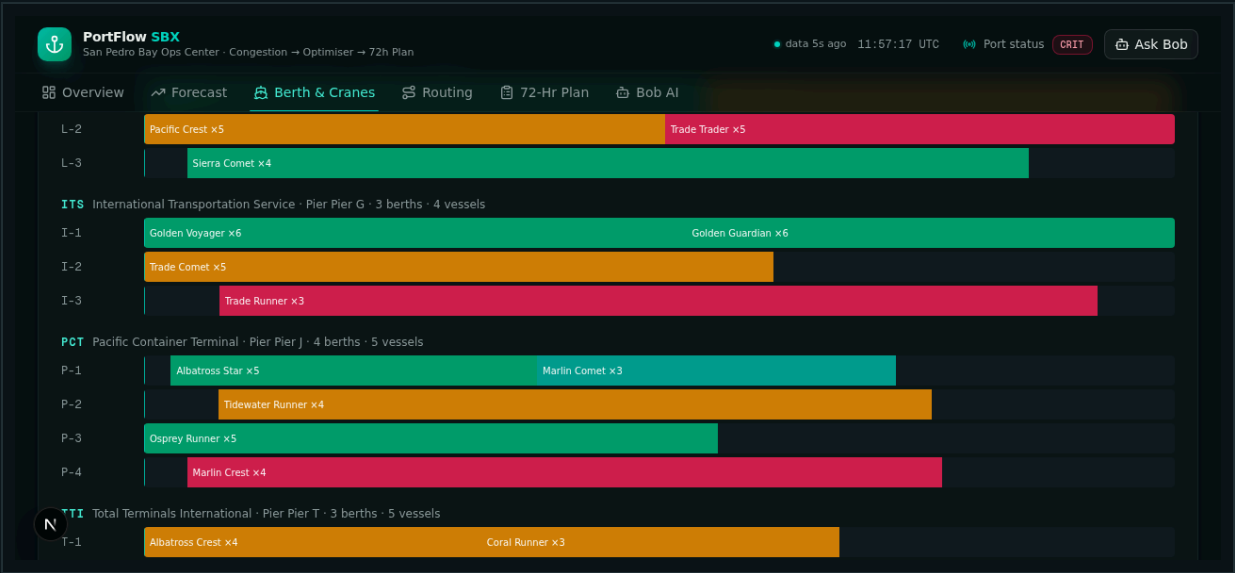
MOVES WORKED

91.1%

BERTH
UTILISATION

- P1 **Priority selection** — weighted score (anchored time, LOA, moves, reefers) picks the serviced set under the 72h horizon.
- P2 **Ready-time sequencing** — FIFO within equal priority; long berths reserved by mismatch penalty (+8 score).
- P3 **Swap + gap-insertion** — interleaved local search packs the schedule; 28 moves/crane-h, ≤8 cranes, 2h buffer.

Deliberately oversubscribed demo (~3× demand) — in crisis the optimiser maximises throughput & priority-weighted fairness, not raw average wait; unslottable vessels become routing candidates. Deltas always report what was gained and lost.



DASHBOARD · BERTH & CRANES TAB — 72H GANTT, FIFO COMPARISON, CRANE DEPLOYMENT

Divert economics, priced before the last leg.

- DIVERT — wait ≥ 48h

SLOW_STEAM — ≥ 18h
- PRIORITY_WINDOW — ≥ 10h

HOLD — monitor

Cost model: **\$32k/day** operating cost, **\$180/unit** reefer spoilage risk, 35% slow-steam fuel saving — every recommendation ships with a ranked savings estimate, confidence and a rationale naming its numbers.

ALTERNATE PORT	DISTANCE	TRANSIT
Oakland	500 nm	33 h
Seattle-Tacoma	1,180 nm	79 h
Prince Rupert	1,260 nm	84 h
Ensenada	150 nm	10 h

\$2.07M

ESTIMATED SAVINGS · LIVE DEMO RUN

38

RECOMMENDATIONS

17

DIVERT CANDIDATES

10

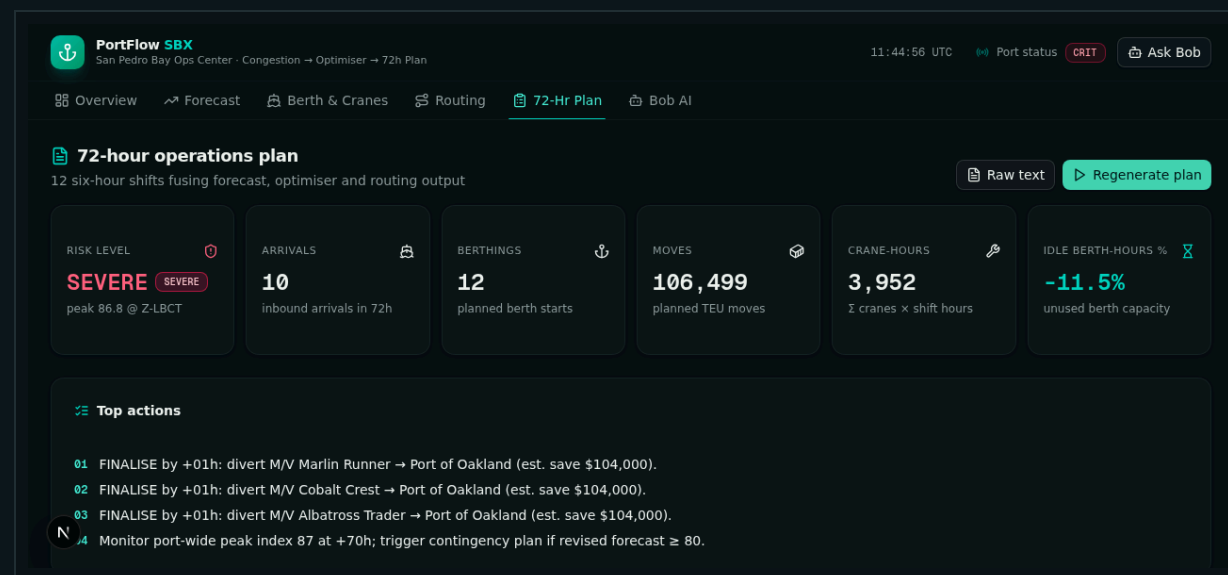
PRIORITY WINDOWS

17 DIVERT · 2 SLOW-STEAM · 10 PRIORITY · 9 HOLD — COMPUTED LIVE FROM FORECAST WAITS

One fused handover plan. Twelve 6-hour shifts.

The plan generator fuses all three engines into a single, shift-by-shift document a supervisor can actually run — JSON for systems, human-readable text for the wall.

- › **Arrivals & berthings** — per shift, with zone and berth, from forecast + optimiser.
- › **Crane deployment** — per terminal, per shift, capped at the 72h horizon.
- › **Congestion alerts** — WATCH / WARN / CRIT with peak index and hour per zone.
- › **Routing actions** — divert/slow-steam orders with decide-by deadlines.
- › **Supervisor checklists** — tickable in the UI; printable raw text with copy & download.

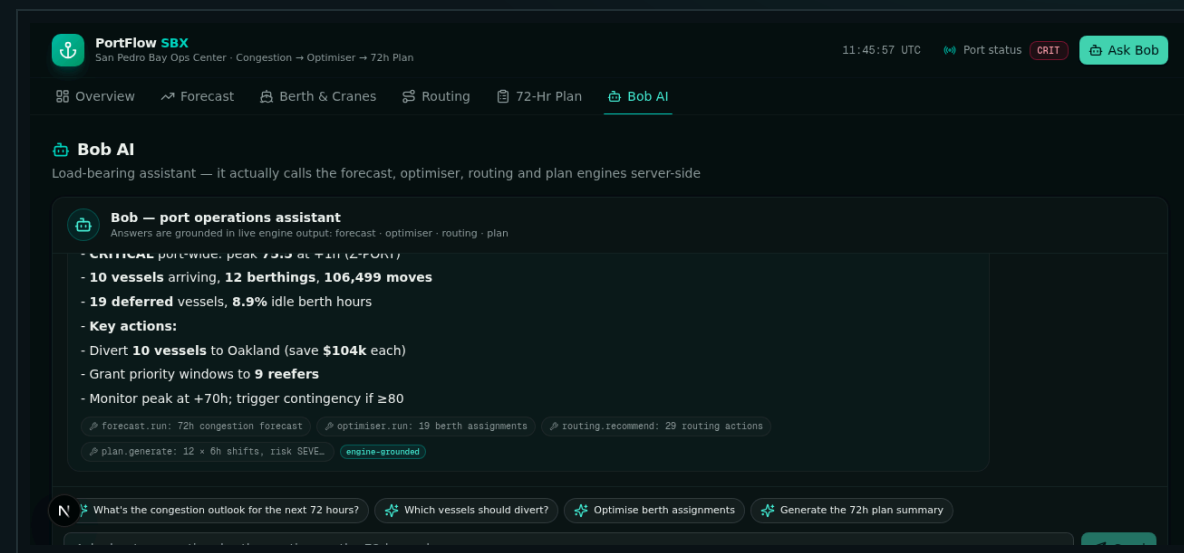


DASHBOARD • 72-HR PLAN TAB — SHIFT TIMELINE, ALERTS, CHECKLISTS, RAW TEXT

A load-bearing assistant — not a chat wrapper.

- 01 **Intent detection** — forecast, berths, routing, plan... or free-form.
- 02 **Real engine calls** — Bob executes the actual pipeline server-side.
- 03 **Grounded prompt** — strictly the engine JSON; no invented numbers allowed.
- 04 **LLM synthesis** — z-ai-web-dev-sdk, backend-only, with tool-action metadata.
- 05 **Deterministic fallback** — if the LLM is unavailable, Bob answers from the engines directly.

Every answer shows an **engine-grounded** or **offline fallback** badge plus the tool chips it ran — the assistant is auditable, exactly like the rest of the system.



DASHBOARD • BOB AI TAB — GROUNDED ANSWER WITH EXECUTED TOOL CHIPS

Every number auditable back to an engine.

72H WARNING

Hotspots seen before they form — not after the queue explodes.

MEASURED DELTAS

Optimiser vs FIFO on real constraints: +10.4% moves, 19 serviced.

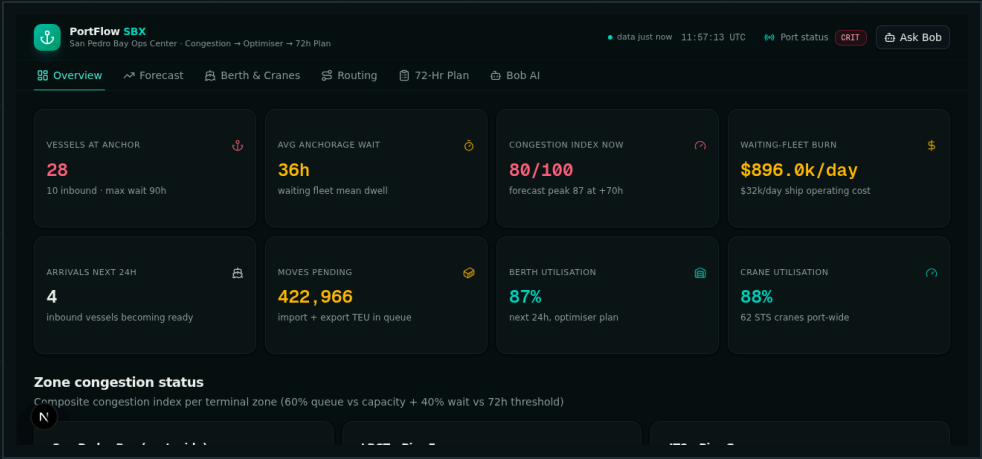
EARLY DIVERT ECONOMICS

\$-priced alternates while slow-steam is still an option.

ONE HANDOVER PLAN

12 shifts, alerts, checklists — fused from all engines.

Demo flow (3–5 min): cold start per docs/setup-guide → Overview → Forecast (zone switch) → Run optimiser → Routing accept/dismiss → Regenerate plan → ask Bob. Honest limits: labelled demo AIS dataset, heuristic optimiser, 13-berth subset.



DASHBOARD · OVERVIEW — THE OPERATOR'S FIRST SCREEN

REPO: README.MD · DOCS/SOLUTION-OVERVIEW.MD · DOCS/ARCHITECTURE.MD
SCREENSHOTS: DEMO/SCREENSHOTS/ · VIDEO: DEMO/DEMO-VIDEO-LINK.TXT